

Model Performance Comparison (Test Set)
Values shown as Mean (95% CI)

Model	AUC	AUPRC	Accuracy	Sensitivity	Specificity	F1	Ppv	Npv	Sens At 90 Spec	Sens At 70 Spec
3D-Inception	0.546 (0.479-0.613)	0.420 (0.351-0.490)	0.390 (0.372-0.408)	0.974 (0.901-1.047)	0.032 (-0.037-0.102)	0.548 (0.533-0.562)	0.381 (0.377-0.386)	0.323 (-0.252-0.898)	0.105 (0.040-0.171)	0.337 (0.203-0.470)
Levit-RI	0.476 (0.450-0.502)	0.369 (0.334-0.404)	0.384 (0.365-0.403)	0.932 (0.843-1.020)	0.048 (-0.025-0.122)	0.534 (0.513-0.555)	0.375 (0.366-0.383)	0.409 (0.117-0.701)	0.058 (-0.013-0.129)	0.253 (0.193-0.312)
Levit-Attention	0.514 (0.496-0.532)	0.395 (0.358-0.433)	0.404 (0.375-0.433)	0.932 (0.829-1.034)	0.081 (-0.026-0.188)	0.542 (0.524-0.560)	0.383 (0.379-0.388)	0.540 (0.158-0.922)	0.079 (0.006-0.152)	0.295 (0.213-0.376)
3D-ResNet-18	0.562 (0.471-0.654)	0.447 (0.329-0.565)	0.444 (0.358-0.530)	0.921 (0.736-1.106)	0.152 (-0.095-0.398)	0.556 (0.531-0.580)	0.403 (0.374-0.433)	0.709 (0.193-1.225)	0.147 (-0.020-0.315)	0.368 (0.172-0.564)
CNN-LSTM	0.637 (0.574-0.701)	0.466 (0.408-0.524)	0.466 (0.372-0.560)	0.921 (0.799-1.043)	0.187 (-0.031-0.405)	0.568 (0.540-0.596)	0.414 (0.372-0.456)	0.883 (0.730-1.036)	0.079 (0.022-0.136)	0.489 (0.325-0.654)
Attention Pool Noinitweights	0.832 (0.806-0.858)	0.764 (0.704-0.823)	0.796 (0.743-0.849)	0.726 (0.614-0.839)	0.839 (0.723-0.955)	0.730 (0.675-0.785)	0.748 (0.645-0.851)	0.836 (0.791-0.881)	0.732 (0.625-0.838)	0.800 (0.782-0.818)
HMV-MIL	0.885 (0.870-0.900)	0.868 (0.843-0.892)	0.794 (0.683-0.905)	0.795 (0.624-0.965)	0.794 (0.558-1.029)	0.750 (0.663-0.837)	0.746 (0.559-0.934)	0.871 (0.804-0.939)	0.774 (0.702-0.845)	0.874 (0.831-0.916)
NoKeyframe	0.878 (0.847-0.908)	0.866 (0.828-0.905)	0.830 (0.800-0.860)	0.732 (0.576-0.887)	0.890 (0.806-0.975)	0.762 (0.698-0.826)	0.823 (0.693-0.954)	0.849 (0.787-0.910)	0.726 (0.629-0.823)	0.868 (0.828-0.908)
NoPathology	0.879 (0.863-0.894)	0.840 (0.806-0.873)	0.826 (0.805-0.847)	0.721 (0.537-0.905)	0.890 (0.782-0.999)	0.754 (0.693-0.815)	0.826 (0.706-0.945)	0.847 (0.771-0.923)	0.784 (0.736-0.833)	0.879 (0.835-0.923)
UniformSampling	0.869 (0.854-0.884)	0.846 (0.822-0.871)	0.812 (0.788-0.836)	0.663 (0.542-0.784)	0.903 (0.843-0.963)	0.726 (0.668-0.783)	0.816 (0.750-0.882)	0.817 (0.770-0.863)	0.726 (0.672-0.781)	0.853 (0.835-0.871)
No RI Full Train	0.830 (0.798-0.863)	0.764 (0.689-0.839)	0.784 (0.750-0.818)	0.584 (0.468-0.701)	0.906 (0.851-0.962)	0.669 (0.591-0.748)	0.800 (0.727-0.874)	0.782 (0.744-0.821)	0.642 (0.551-0.733)	0.800 (0.732-0.868)
Original Noinitweights	0.837 (0.811-0.862)	0.773 (0.720-0.827)	0.808 (0.788-0.828)	0.747 (0.577-0.918)	0.845 (0.767-0.923)	0.742 (0.674-0.810)	0.759 (0.680-0.837)	0.852 (0.781-0.922)	0.684 (0.615-0.754)	0.821 (0.758-0.884)
Original	0.842 (0.827-0.858)	0.756 (0.691-0.820)	0.788 (0.724-0.852)	0.711 (0.532-0.889)	0.835 (0.639-1.032)	0.718 (0.675-0.761)	0.771 (0.596-0.947)	0.834 (0.771-0.898)	0.674 (0.559-0.788)	0.837 (0.801-0.873)
Video Transformer	0.740 (0.721-0.760)	0.618 (0.589-0.648)	0.614 (0.516-0.712)	0.732 (0.613-0.850)	0.542 (0.322-0.762)	0.592 (0.556-0.629)	0.510 (0.411-0.610)	0.766 (0.726-0.807)	0.395 (0.343-0.446)	0.663 (0.627-0.699)